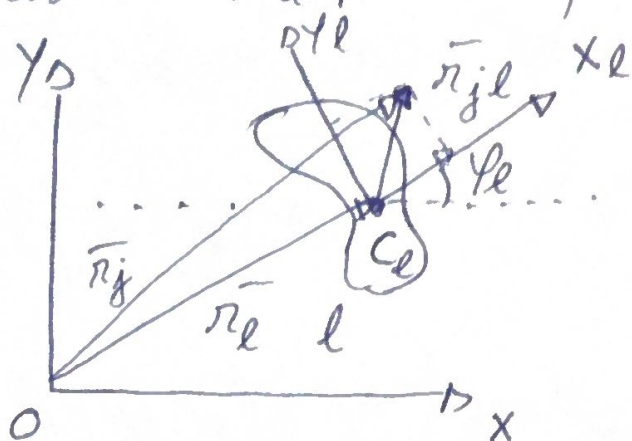


Considerăm un sistem de n corpuri, fiecare fiind compus de un număr arbitrar de puncte aflate în mișcare plan-paralelă.



Fiecare corp are un sistem de coordonate x_e, y_e , aflat în centrul său de masă. Sistemul este considerat aflat la un unghi ψ_e ce corespunde mișcării corpului. Versorii sistemului sunt $\bar{i}_e, \bar{j}_e$ neconstanți.

$$\bar{r}_{je} = d_{xje} \bar{i}_e + d_{yje} \bar{j}_e$$

$$\bar{i}_e = \cos \psi_e \bar{i} + \sin \psi_e \bar{j}$$

$$\bar{j}_e = \sin \psi_e \bar{i} + \cos \psi_e \bar{j}$$

$$\Rightarrow \bar{r}_{je} = (d_{xje} \cos \psi_e + d_{yje} \sin \psi_e) \bar{i} + (-d_{xje} \sin \psi_e + d_{yje} \cos \psi_e) \bar{j}$$

$$\bar{r}_j = \bar{r}_e + \bar{r}_{je} = \begin{bmatrix} x_e + d_{xje} \cos \psi_e + d_{yje} \sin \psi_e \\ y_e + d_{xje} \sin \psi_e + d_{yje} \cos \psi_e \end{bmatrix}$$

$$\bar{v}_j = \dot{\bar{r}}_j = \begin{bmatrix} \dot{x}_e + \dot{\psi}_e (-d_{xje} \sin \psi_e + d_{yje} \cos \psi_e) \\ \dot{y}_e + \dot{\psi}_e (d_{xje} \cos \psi_e + d_{yje} \sin \psi_e) \end{bmatrix} = \begin{bmatrix} v_{jx} \\ v_{jy} \end{bmatrix}$$

$$v_j^2 = \langle \bar{v}_j, \bar{v}_j \rangle = v_{jx}^2 + v_{jy}^2 = \dot{x}_e^2 + 2\dot{x}_e\dot{\psi}_e(-d_{xje} \sin \psi_e + d_{yje} \cos \psi_e) + \dot{\psi}_e^2 (d_{xje}^2 \sin^2 \psi_e + d_{yje}^2 \cos^2 \psi_e + d_{xje}^2 \cos^2 \psi_e + d_{yje}^2 \sin^2 \psi_e)$$

$$= \dot{x}_e^2 + \dot{y}_e^2 + \dot{\varphi}_e^2 (d_{xje}^2 \sin^2 \varphi_e + d_{xje}^2 \cos^2 \varphi_e + d_{yje}^2 \cos^2 \varphi_e + d_{yje}^2 \sin^2 \varphi_e - 2d_{xje} \sin \varphi_e d_{yje} \cos \varphi_e + 2d_{xje} \cos \varphi_e d_{yje} \sin \varphi_e) + 2\dot{\varphi}_e [\dot{x}_e (-d_{xje} \sin \varphi_e + d_{yje} \cos \varphi_e) + \dot{y}_e (-d_{xje} \cos \varphi_e - d_{yje} \sin \varphi_e)]$$

$$= \dot{x}_e^2 + \dot{y}_e^2 + \dot{\varphi}_e^2 (d_{xje}^2 + d_{yje}^2) + 2\dot{\varphi}_e [d_{xje} (-\dot{x}_e \sin \varphi_e - \dot{y}_e \cos \varphi_e) + d_{yje} (\dot{x}_e \cos \varphi_e - \dot{y}_e \sin \varphi_e)]$$

$$E_e = \sum_{j=1}^N E_j = \frac{1}{2} \sum_{j=1}^N m_j V^2 = \frac{1}{2} \left(\sum_{j=1}^N m_j \dot{x}_e^2 + \sum_{j=1}^N m_j \dot{y}_e^2 + \sum_{j=1}^N \dot{\varphi}_e^2 m_j (d_{xje}^2 + d_{yje}^2) + \sum_{j=1}^N m_j d_{xje} \cdot 2\dot{\varphi}_e (-\dot{x}_e \sin \varphi_e - \dot{y}_e \cos \varphi_e) + \sum_{j=1}^N m_j d_{yje} \cdot 2\dot{\varphi}_e (\dot{x}_e \cos \varphi_e - \dot{y}_e \sin \varphi_e) \right)$$

$$E_e = \sum_{j=1}^N m_j = M_e, \quad \sum_{j=1}^N m_j d_{xje} = S_y, \quad \sum_{j=1}^N m_j d_{yje} = S_x, \quad \sum_{j=1}^N m_j (d_{xje}^2 + d_{yje}^2) = J_{ze}$$

Ne aflăm în planul xOy , deci $S_x = S_y = 0$ întrucât sistemul de coordonate are originea în centrul de masă, deci $S_x = S_y = 0$

$$\Rightarrow E_e = \frac{1}{2} \dot{x}_e^2 M_e + \frac{1}{2} \dot{y}_e^2 M_e + \frac{1}{2} \dot{\varphi}_e^2 J_{ze}$$

În planul xOy , avem un singur moment de inerție planar: $\begin{cases} J_{xx}^0 \\ J_{yy}^0 \\ J_{zz}^0 \end{cases}$
 8.7 Considerăm vectorul de coordonate generalizate:

$$q = \begin{bmatrix} x_1 \\ y_1 \\ \varphi_1 \\ \vdots \\ x_n \\ y_n \\ \varphi_n \end{bmatrix} \Rightarrow \dot{q} = \begin{bmatrix} \dot{x}_1 \\ \dot{y}_1 \\ \dot{\varphi}_1 \\ \vdots \\ \dot{x}_n \\ \dot{y}_n \\ \dot{\varphi}_n \end{bmatrix}$$

Energia intrinsecă sistem este:

$$E = \sum_{l=1}^n E_l = \frac{1}{2} \sum_{l=1}^n \dot{x}_l^2 M_l + \dot{y}_l^2 M_l + \dot{\varphi}_l^2 J_l$$

$$= \frac{1}{2} \cdot \dot{x}_1^2 M_1 + \dot{y}_1^2 M_1 + \dot{\varphi}_1^2 J_1 + \dot{x}_2^2 M_2 + \dot{y}_2^2 M_2 + \dot{\varphi}_2^2 J_2 + \dots + \dot{x}_n^2 M_n + \dot{y}_n^2 M_n + \dot{\varphi}_n^2 J_n$$

$$= \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n a_{ij} \dot{q}_i \dot{q}_j \quad \text{este o formă pătratică în } \dot{q}, \text{ având}$$

matricea diagonală:

$$A = \begin{pmatrix} M_1 & 0 & 0 & 0 & \dots & 0 \\ 0 & M_1 & 0 & 0 & \dots & 0 \\ 0 & 0 & J_1 & 0 & \dots & 0 \\ 0 & 0 & 0 & M_2 & \dots & 0 \\ 0 & 0 & 0 & 0 & M_2 & J_2 \\ 0 & 0 & 0 & 0 & 0 & \ddots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & M_n \\ 0 & 0 & \dots & \dots & \dots & 0 & M_n \\ 0 & 0 & \dots & \dots & \dots & 0 & 0 & J_n \\ 0 & 0 & \dots & \dots & \dots & 0 & 0 & 0 \end{pmatrix}$$

$$E = \frac{1}{2} \dot{q}^T A \dot{q}$$

Ecuațiile lui Lagrange în cazul sistemelor cu variabile independente

$$\text{Fie: } \begin{cases} f_1(q_1, \dots, q_n, t) = 0 \\ \vdots \\ f_p(q_1, \dots, q_n, t) = 0 \end{cases}$$

legăturile dependente a sistemului

$$\text{Notăm } F(q, t) = (f_1(q, t), f_2(q, t), \dots, f_p(q, t))$$

$$\text{unde } q = \begin{pmatrix} q_1 \\ \vdots \\ q_n \end{pmatrix} \Rightarrow \text{Putem scrie legăturile}$$

ca: $F(q, t) = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$

Ecuațiile lui Lagrange au formă:

$$\frac{d}{dt} \left(\frac{\partial \tilde{T}}{\partial \dot{q}_k} \right) - \frac{\partial \tilde{T}}{\partial q_k} = Q_k + \sum_{j=1}^p \lambda_j \frac{\partial f_j}{\partial q_k}, \text{ unde } \lambda_j \text{ sunt}$$

multipliatorii lui
Lagrange

Calculăm Jacobianul lui F :

$$J_F = \begin{bmatrix} \frac{\partial f_1}{\partial q_1} & \frac{\partial f_1}{\partial q_2} & \dots & \frac{\partial f_1}{\partial q_{3n}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_p}{\partial q_1} & \frac{\partial f_p}{\partial q_2} & \dots & \frac{\partial f_p}{\partial q_{3n}} \end{bmatrix}$$

$$\text{Fie } \Delta = \begin{bmatrix} \lambda_1 \\ \vdots \\ \lambda_p \end{bmatrix}$$

Observăm că atunci când calculăm $J_F^T \Delta$, obținem:

$$\begin{bmatrix} \frac{\partial f_1}{\partial q_1} & \dots & \frac{\partial f_p}{\partial q_1} \\ \frac{\partial f_1}{\partial q_2} & \dots & \frac{\partial f_p}{\partial q_2} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_1}{\partial q_{3n}} & \dots & \frac{\partial f_p}{\partial q_{3n}} \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \vdots \\ \lambda_p \end{bmatrix} = \begin{bmatrix} \sum_{j=1}^p \lambda_j \frac{\partial f_j}{\partial q_1} \\ \sum_{j=1}^p \lambda_j \frac{\partial f_j}{\partial q_2} \\ \vdots \\ \sum_{j=1}^p \lambda_j \frac{\partial f_j}{\partial q_{3n}} \end{bmatrix}$$

Vectorul obținut are chiar termenii din ec. lui Lagrange.

Când scriem ecuațiile lui Lagrange obținem:

$$\left\{ \begin{aligned} \frac{d}{dt} \left(\frac{\partial \bar{E}}{\partial \dot{q}_1} \right) - \frac{\partial \bar{E}}{\partial q_1} &= Q_1 + \sum_{j=1}^p \lambda_j \frac{\partial f_j}{\partial q_1} \\ \frac{d}{dt} \left(\frac{\partial \bar{E}}{\partial \dot{q}_2} \right) - \frac{\partial \bar{E}}{\partial q_2} &= Q_2 + \sum_{j=1}^p \lambda_j \frac{\partial f_j}{\partial q_2} \\ &\vdots \\ \frac{d}{dt} \left(\frac{\partial \bar{E}}{\partial \dot{q}_{3n}} \right) - \frac{\partial \bar{E}}{\partial q_{3n}} &= Q_{3n} + \sum_{j=1}^p \lambda_j \frac{\partial f_j}{\partial q_{3n}} \end{aligned} \right.$$

$$\Rightarrow \frac{d}{dt} \left(\frac{\partial \bar{E}}{\partial \dot{q}} \right) - 0 = Q + J_f^T \Delta$$

$$(-) \quad A \ddot{q} = Q + J_f^T \Delta$$

Când scriem, în $F=0$